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Álgebra Lineal

Grupo 17

Semestre 2022-2

Serie 3
Tema 4Sea el producto interno en $\mathbb{C}^2$ definido por:

$$(\bar{v} | \bar{w}) = v_1 \bar{w}_1 + 2v_2 \bar{w}_2 \quad \forall \bar{v} = (v_1, v_2), \bar{w} = (w_1, w_2) \in \mathbb{C}^2$$

Donde $\bar{w}_1$ y $\bar{w}_2$ son el conjugado de w_1 y w_2 respectivamente. Para los vectores $\bar{v} = (i, 2-i)$ y $\bar{w} = (-1, 1+i)$.

a) Obtener el coseno del ángulo entre los vectores $\bar{v}$ y $\bar{w}$.

b) Calcular la distancia del vector $\bar{v}$ al vector $\bar{w}$.

c) Determinar si $\bar{v}$ y $\bar{w}$ son ortogonales.

Solución a)

$$\cos \theta = \frac{(\bar{v} | \bar{w})}{\|\bar{v}\| \|\bar{w}\|}$$

$$(\bar{v} | \bar{w}) = \begin{pmatrix} i & 2-i \end{pmatrix} \begin{pmatrix} -1 & 1+i \end{pmatrix}$$

$\begin{matrix} v_1 & v_2 & w_1 & w_2 \end{matrix}$

$$(\bar{v} | \bar{w}) = v_1 \bar{w}_1 + 2v_2 \bar{w}_2$$

$$(\bar{v} | \bar{w}) = (i)(-1) + 2(2-i)(1+i)$$

$$(\bar{v} | \bar{w}) = (\bar{a})(-1) + 2(2-\bar{a})(1-\bar{a})$$

$$(\bar{v} | \bar{w}) = -\bar{a} + 2 - 6\bar{a}$$

$$(\bar{v} | \bar{w}) = \underline{2-7\bar{a}} //$$

$$||\bar{v}|| = ||(\bar{a}, 2-\bar{a})||$$

$$= [(\bar{a}, 2-\bar{a}) | (\bar{a}, 2-\bar{a})]^{1/2}$$

$$= [(\bar{a})(\bar{a}) + 2(2-\bar{a})(2-\bar{a})]^{1/2}$$

$$= [(\bar{a})(-\bar{a}) + (4-2\bar{a})(2+\bar{a})]^{1/2}$$

$$= [-\bar{a}^2 + 8 + 4\bar{a} - 4\bar{a} - 2\bar{a}^2]^{1/2}$$

$$= (8-3\bar{a}^2)^{1/2}$$

$$||\bar{v}|| = \sqrt{11} //$$

$$||\bar{w}|| = ||(-1, 1+\bar{a})||$$

$$= [(-1, 1+\bar{a}) | (-1, 1+\bar{a})]^{1/2}$$

$$= [(-1)(-1) + 2(1+\bar{a})(1+\bar{a})]^{1/2}$$

$$= [(-1)(-1) + (2+2\bar{a})(1-\bar{a})]^{1/2}$$

$$= (1+2-2\bar{a}+2\bar{a}-2\bar{a}^2)^{1/2}$$

$$= (3-2\bar{a}^2)$$

$$||\bar{w}|| = \sqrt{5} //$$

$$\cos \theta = \frac{2-7\bar{a}}{(\sqrt{11})(\sqrt{5})}$$

$$\therefore \theta = \arccos \left(\frac{2-7i}{\sqrt{55}} \right) \approx 87.9^\circ$$

74.35°

b)

$$d(\bar{v}, \bar{w}) = ||\bar{w} - \bar{v}||$$

$$d(\bar{v}, \bar{w}) = ||(-1, 1+i) - (i, 2-i)||$$

$$d(\bar{v}, \bar{w}) = ||(-1-i, -1+2i)||$$

$$(\bar{v} | \bar{w}) = v_1 \bar{w}_1 + 2v_2 \bar{w}_2$$

$$\begin{aligned} ||(\bar{v} | \bar{w})|| &= [(-1-i)(-1+2i) + 2(-1)(-1+2i)]^{1/2} \\ &= [(-1-i)(-1+2i) + 2(-1)(-1+2i)]^{1/2} \\ &= [(-1-i)(-1+2i) + (-2+4i)(-1+2i)]^{1/2} \\ &= (1-i+2i-i^2 + 2-4i+2i^2-8i^2)^{1/2} \\ &= (3-9i^2)^{1/2} \end{aligned}$$

$$||\bar{v} \bar{w}|| = \sqrt{12}$$

$\therefore$ La distancia entre $\bar{v}$ y $\bar{w}$ es de $\sqrt{12}$

c)

$$(\bar{v} | \bar{w}) = v_1 \bar{w}_1 + 2v_2 \bar{w}_2$$

$$\bar{v} = (i, 2-i)$$

$$\bar{w} = (-1, 1+i)$$

$$\begin{aligned} (\bar{v} | \bar{w}) &= (i, 2-i) | (-1, 1+i) \\ &= (i)(-1) + 2(2-i)(1+i) \\ &= (-i + 2(2-i)(1+i)) \end{aligned}$$

Nombre:

Día

Mes

Año

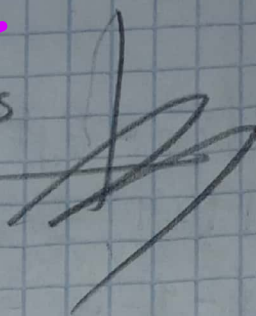
Folio

Tema:

$$= 1i + 2 - 6i$$

$$(\bar{v} | \bar{w}) = 2 - 7i$$

$\therefore \bar{v}$ y $\bar{w}$ no son ortogonales



Sea el espacio vectorial $\mathbb{R}^3$ con producto interno definido por:

$$(\bar{x} | \bar{y}) = 3x_1y_1 + x_2y_2 + 2x_3y_3$$

$$\bar{x} = (x_1, x_2, x_3), \bar{y} = (y_1, y_2, y_3) \in \mathbb{R}^3$$

y sea $W = \{(x, y, y-2x) \mid x, y \in \mathbb{R}\}$ un subespacio de $\mathbb{R}^3$. Determinar:

a) Una base ortogonal de W .

b) La proyección del vector $\bar{a} = (1, 3, -1)$ sobre W

c) La distancia entre el vector $\bar{a}$ y su proyección sobre W .

Solución

a) $W = (x, y, y-2x)$

$$\left. \begin{array}{l} x=1 \\ y=0 \end{array} \right\} (1, 0, -2)$$

$$\left. \begin{array}{l} x=0 \\ y=1 \end{array} \right\} (0, 1, 1)$$

$$B = \left\{ \underbrace{(1, 0, -2)}_{\bar{v}_1}, \underbrace{(0, 1, 1)}_{\bar{v}_2} \right\}$$

Nombre:

Día

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Tema:

$$\vec{w}_1 = \vec{v}_1$$

$$\vec{w}_1 = (1, 0, -2) //$$

$$\vec{w}_2 = \vec{v}_2 - \frac{(\vec{v}_2 | \vec{w}_1)}{(\vec{w}_1 | \vec{w}_1)} \vec{w}_1$$

$$\vec{w}_2 = (0, 1, 1) - \frac{[(0, 1, 1) | (1, 0, -2)]}{[(1, 0, -2) | (1, 0, -2)]} (1, 0, -2)$$

$$(\vec{x} | \vec{y}) = 3x_1y_1 + x_2y_2 + 2x_3y_3$$

$$\begin{array}{ccc|ccc} (0 & 1 & 1) & | & (1 & 0 & -2) \\ x_1 & x_2 & x_3 & & y_1 & y_2 & y_3 \end{array} = 3(0)(1) + (1)(0) + 2(1)(-2)$$

$$= 0 + 0 - 4$$

$$= -4 //$$

$$\begin{array}{ccc|ccc} (1 & 0 & -2) & | & (1 & 0 & -2) \\ x_1 & x_2 & x_3 & & y_1 & y_2 & y_3 \end{array} = 3(1)(1) + (0)(0) + 2(-2)(-2)$$

$$= 3 + 0 + 8$$

$$= 11 //$$

$$\vec{w}_2 = (0, 1, 1) - \left(\frac{-4}{11} \right) (1, 0, -2)$$

$$\vec{w}_2 = (0, 1, 1) - (-4/11, 0, 8/11)$$

$$\vec{w}_2 = (4/11, 1, 3/11)$$

$$\therefore \text{Bortg} = \{(1, 0, -2), (4/11, 1, 3/11)\}$$

b)

$$\begin{aligned} \|(1, 0, -2)\| &= \left[\underset{x_1}{1} \underset{x_2}{0} \underset{x_3}{-2} \mid \underset{y_1}{1} \underset{y_2}{0} \underset{y_3}{-2} \right]^{1/2} \\ &= [3(1)(1) + (0)(0) + 2(-2)(-2)]^{1/2} \\ &= (3 + 0 + 8)^{1/2} \end{aligned}$$

$$\|(1, 0, -2)\| = \sqrt{11}$$

$$\begin{aligned} \|(4/11, 1, 3/11)\| &= \left[\underset{x_1}{4/11} \underset{x_2}{1} \underset{x_3}{3/11} \mid \underset{y_1}{4/11} \underset{y_2}{1} \underset{y_3}{3/11} \right]^{1/2} \\ &= \left[3\left(\frac{4}{11}\right)\left(\frac{4}{11}\right) + (1)(1) + 2\left(\frac{3}{11}\right)\left(\frac{3}{11}\right) \right]^{1/2} \\ &= \left(\frac{48}{121} + 1 + \frac{18}{121} \right)^{1/2} \end{aligned}$$

$$\|(4/11, 1, 3/11)\| = \sqrt{\frac{187}{121}} = \frac{\sqrt{187}}{11}$$

$$\text{Borton} = \left\{ \frac{1}{\sqrt{11}}(1, 0, -2), \frac{1}{\sqrt{187}}\left(\frac{4}{11}, 1, \frac{3}{11}\right) \right\}$$

$$\sum_{i=1}^n (\bar{a} | \bar{e}_i) \bar{e}_i \rightarrow (\bar{a} | \bar{e}_1) \bar{e}_1 + (\bar{a} | \bar{e}_2) \bar{e}_2$$

$$\begin{aligned} \sum_{i=1}^n ((1, 3, -1) | \bar{e}_i) \bar{e}_i &= ((1, 3, -1) | \frac{1}{\sqrt{11}}(1, 0, -2)) \frac{1}{\sqrt{11}}(1, 0, -2) \\ &\quad + ((1, 3, -1) | \frac{1}{\sqrt{187}}(\frac{4}{11}, 1, \frac{3}{11})) \frac{1}{\sqrt{187}}(\frac{4}{11}, 1, \frac{3}{11}) \end{aligned}$$

$$(\bar{a} | \bar{e}_1) \bar{e}_1$$

$$\begin{aligned} \left[\underset{x_1}{1} \underset{x_2}{3} \underset{x_3}{-1} \mid \underset{y_1}{1/\sqrt{11}} \underset{y_2}{0} \underset{y_3}{-2/\sqrt{11}} \right] &= 3(1)\left(\frac{1}{\sqrt{11}}\right) + (3)(0) + 2(-1)\left(-\frac{2}{\sqrt{11}}\right) \\ &= \frac{3}{\sqrt{11}} + 0 + \frac{4}{\sqrt{11}} \end{aligned}$$

$$= \frac{7}{\sqrt{11}}$$

$$(\bar{a} | \bar{e}_1) \bar{e}_1 = \left(\frac{7}{\sqrt{11}} \right) \left(\frac{1}{\sqrt{11}}, 0, -\frac{2}{\sqrt{11}} \right)$$

$$(\bar{a} | \bar{e}_1) \bar{e}_1 = \left(\frac{7}{\sqrt{11}}, 0, -\frac{14}{\sqrt{11}} \right) //$$

$$(\bar{a} | \bar{e}_2) \bar{e}_2$$

$$\left[(1, 3, -1) \left| \frac{1}{\sqrt{17}} \left(\frac{4}{11}, 1, \frac{3}{11} \right) \right. \right]$$

$$\left[\left(\frac{1}{x_1}, \frac{3}{x_2}, -\frac{1}{x_3} \right) \left| \left(\frac{4}{11\sqrt{17}}, \frac{1}{\sqrt{17}}, \frac{3}{11\sqrt{17}} \right) \right. \right]$$

$$= 3(1) \left(\frac{4}{11\sqrt{17}} \right) + 2(-1) \left(\frac{3}{11\sqrt{17}} \right)$$

$$= \frac{12}{11\sqrt{17}} + \frac{3}{\sqrt{17}} - \frac{6}{11\sqrt{17}}$$

$$= \frac{39\sqrt{17}/11}{17}$$

$$(\bar{a} | \bar{e}_2) \bar{e}_2 = \left(\frac{39\sqrt{17}/11}{17} \right) \left(\frac{4}{11\sqrt{17}}, \frac{1}{\sqrt{17}}, \frac{3}{11\sqrt{17}} \right)$$

$$(\bar{a} | \bar{e}_2) \bar{e}_2 = \left(\frac{156}{187}, \frac{39}{17}, \frac{117}{187} \right) //$$

$$\sum_{i=1}^n (\bar{a} | \bar{e}_i) \bar{e}_i = \left(\frac{7}{11}, 0, -\frac{14}{11} \right) + \left(\frac{156}{187}, \frac{39}{17}, \frac{117}{187} \right)$$

$$\therefore \sum_{i=1}^n (\bar{a} | \bar{e}_i) \bar{e}_i = \left(\frac{25}{17}, \frac{39}{17}, -\frac{11}{17} \right)$$

e)

$$d(\bar{u}, \bar{v}) = \|\bar{v} - \bar{u}\|$$

$$d(\bar{a}, \bar{w}) = \|\bar{w} - \bar{a}\|$$

$$d(\bar{a}, \bar{w}) = \left\| \left(\frac{25}{17}, \frac{39}{17}, -\frac{11}{17} \right) - (1, 3, -2) \right\|$$

$$d(\bar{a}, \bar{w}) = \left\| \left(\frac{8}{17}, -\frac{12}{17}, \frac{6}{17} \right) \right\|$$

$$\begin{aligned} & \left[\left(\frac{8}{17}, -\frac{12}{17}, \frac{6}{17} \right) \cdot \left(\frac{8}{17}, -\frac{12}{17}, \frac{6}{17} \right) \right]^{1/2} \\ &= 3 \left(\frac{8}{17} \right) \left(\frac{8}{17} \right) + \left(-\frac{12}{17} \right) \left(-\frac{12}{17} \right) + 2 \left(\frac{6}{17} \right) \left(\frac{6}{17} \right) \\ &= \frac{192}{289} + \frac{144}{289} + \frac{72}{289} \\ &= \sqrt{\frac{408}{289}} = \sqrt{\frac{24}{17}} \end{aligned}$$

$\therefore$ La distancia entre $\bar{a}$ y el vector $\bar{w}$ es de $\sqrt{\frac{24}{17}}$ u

Sea $\mathcal{P}_1$ el espacio vectorial real de los polinomios de grado menor o igual a uno con coeficientes reales y M el subespacio de $\mathcal{P}_1$ generado por el conjunto

$$A = \left\{ 1-x, -\frac{1}{2} - \frac{1}{2}x \right\}$$

y sea el producto interno definido por

$$(p|q) = \sum_{i=0}^1 2a_i b_i \quad \forall p(x) = a_0 + a_1 x, q(x) = b_0 + b_1 x \in \mathcal{P}_1$$

Determinar el polinomio $m(x) \in M$ más próximo a $r(x) = 1 + 2x$

Solución

$$A = \left\{ 1-x, -\frac{1}{2} - \frac{1}{2}x \right\}$$

$$(p|q) = \sum_{i=0}^1 2a_i b_i =$$

$$(p|q) = 2a_0 b_0 + 2a_1 b_1$$

$$\|(1-x)\| = [(1-x)|(1-x)]^{1/2}$$

$$= [2(1)(1) + 2(-1)(-1)]^{1/2}$$

$$= (2+2)^{1/2} = \sqrt{4} = 2$$

$$(p|q) = 2a_0 b_0 + 2a_1 b_1 x$$

$$b_1 = \frac{\overline{\omega_1}}{\|\overline{\omega_1}\|}$$

$$\|(-\frac{1}{2} - \frac{1}{2}x)\| = [(-\frac{1}{2} - \frac{1}{2}x)|(-\frac{1}{2} - \frac{1}{2}x)]^{1/2}$$

$$= [2(-\frac{1}{2})(-\frac{1}{2}) + 2(-\frac{1}{2})(-\frac{1}{2})]^{1/2}$$

$$= (\frac{1}{2} + \frac{1}{2})^{1/2}$$

$$= \sqrt{1} = 1$$

$$b_2 = \frac{\overline{\omega_2}}{\|\overline{\omega_2}\|}$$

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Tema:

$$B_{ortn} = \left\{ \underbrace{\frac{1}{2}(1-x)}_{b_1}, \underbrace{\left(-\frac{1}{2} - \frac{1}{2}x\right)}_{b_2} \right\}$$

$$\sum_{i=1}^n (\bar{v} | \bar{b}_i) \bar{b}_i$$

$$\sum_{i=1}^2 (\bar{v} | \bar{b}_i) \bar{b}_i = (\bar{v} | \bar{b}_1) \bar{b}_1 + (\bar{v} | \bar{b}_2) \bar{b}_2$$

$$(\bar{v} | \bar{b}_1) \bar{b}_1$$

$$\begin{aligned} \left[\left(\frac{1}{x_1} + 2x_2 \right) \middle| \left(\frac{1}{x_1} - \frac{1}{2}x_2 \right) \right] &= 2(1)\left(\frac{1}{2}\right) + 2(2)\left(-\frac{1}{2}\right) \\ &= 1 - 2 \\ &= -1 \end{aligned}$$

$$(\bar{v} | \bar{b}_1) \bar{b}_1 = (-1)\left(\frac{1}{2}\right)(1-x)$$

$$(\bar{v} | \bar{b}_1) \bar{b}_1 = -\frac{1}{2} + \frac{1}{2}x$$

$$(\bar{v} | \bar{b}_2) \bar{b}_2$$

$$\begin{aligned} \left[(1+2x) \middle| \left(-\frac{1}{2} - \frac{1}{2}x\right) \right] &= 2(1)\left(-\frac{1}{2}\right) + 2(2)\left(-\frac{1}{2}\right) \\ &= -1 - 2 \\ &= -3 \end{aligned}$$

$$(\bar{v} | \bar{b}_2) \bar{b}_2 = (-3)\left(-\frac{1}{2} - \frac{1}{2}x\right)$$

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Tema:

$$(\bar{v} | b_2) b_2 = \frac{3}{2} + \frac{3}{2}x$$

$$\sum_{i=1}^2 (\bar{v} | b_i) b_i = \left(-\frac{1}{2} + \frac{1}{2}x\right) \cdot \left(\frac{3}{2} + \frac{3}{2}x\right)$$

$$\therefore \sum_{i=1}^2 (\bar{v} | b_i) b_i = 1 + 2x$$

Sean el espacio vectorial $\mathbb{R}^3$ con producto interno usual y

$$W = \{(x, y, z) \mid x - y = 0; x, y, z \in \mathbb{R}\}$$

un subespacio de $\mathbb{R}^3$.

a) Determinar el complemento ortogonal $W^\perp$ de W

b) Expresar el vector $\vec{v} = (1, 0, 1)$ como la suma $\vec{v} = \vec{a} + \vec{b}$ donde $\vec{a} \in W$ y $\vec{b} \in W^\perp$

Solución de a)

$$W = \{(x, x, z) \mid x, z \in \mathbb{R}\}$$

$$\left. \begin{array}{l} x=1 \\ z=0 \end{array} \right\} (1, 1, 0)$$

$$\left. \begin{array}{l} x=0 \\ z=1 \end{array} \right\} (0, 0, 1)$$

$$B = \{(1, 1, 0), (0, 0, 1)\}$$

$$W^\perp = \{\vec{v} \in V \mid (\vec{v} \mid \vec{w}) = 0; \vec{w} \in W\}$$

$$W^\perp = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y, z) \mid (1, 1, 0) = 0, (x, y, z) \mid (0, 0, 1) = 0\}$$

$$\begin{aligned} [(x, y, z) \mid (1, 1, 0)] &= x + y + 0 \\ &= x + y = 0 \end{aligned}$$

$$\begin{aligned} y &= -x \\ x &= x \end{aligned}$$

Nombre:

Día

Mes

Año

Folio

Tema:

$$[(x, y, z) | (0, 0, 1)] = 0 + 0 + z \\ = z = 0$$

$$\therefore w^+ = \{(x, -x, 0) | x \in \mathbb{R}\}$$

b)

$$\bar{v} = \bar{a} + \bar{b} \Leftrightarrow \bar{v} = \bar{w} + \bar{w}^\perp$$

$$(1, 0, 1) = (x, x, z) + (y, -x, 0)$$

$$(1, 0, 1) = (x+x, x-x, z)$$

$$(1, 0, 1) = (2x, 0, z)$$

$$2x = 1 \quad z = 1 \\ x = \frac{1}{2}$$

$$\therefore (1, 0, 1) = \left(\frac{1}{2}, \frac{1}{2}, 1\right) + \left(\frac{1}{2}, -\frac{1}{2}, 0\right)$$

Comprobación

$$(1, 0, 1) = (x+y, x-x, z)$$

$$(1, 0, 1) = \left(\frac{1}{2} + \frac{1}{2}, \frac{1}{2} - \frac{1}{2}, 1\right)$$

$$(1, 0, 1) = (1, 0, 1)$$